

Jongseo Choi

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Education

Technical University of Chemnitz (Germany)

Oct 2018 – Sep 2020

M.Sc. Automotive Software Engineering

- Implementation of a communication system with Raspberry Pi and Arduino
- **Courses:** Design SW for Embedded Systems, machine learning, multi-core system, Real-time system, Dependable system, AUTOSAR, etc.

Chungbuk National University (S.Korea)

Mar 2008 – Jan 2015

B.Sc. Electronic Engineering

(including 2 years military service)

- Development of a line tracer robot using an infrared sensor, a DC motor, and an Atmega microcontroller
- Development of a mini-wheelchair controlled by a wireless helmet using an accelerometer and gyro sensor

Experience

Senior Engineering Manager (Hyundai Motor Company, S.Korea)

Mar 2025 – Present

- Robot Application Development: Designed and implemented SPOT-based solutions for safety patrol, machine inspection, and quality inspection with spot arm manipulation.
- Defined Concepts of Operation (ConOps) and technical requirements for integrating robots (e.g., Spot, Atlas) into factory environments.
- Defined Technical Requirements and established R&D Roadmaps for humanoid robot integration.
- Humanoid R&D Research: Researched cutting-edge humanoid technologies, including world model integration, simulation, teleoperation, and VLA.

Senior SW Engineer (ThorDrive, S.Korea)

Nov 2020 – Mar 2025

- Development of Autonomous Shuttle/Taxi Solutions
- Development of Automated Parking Path Planning Solutions
- Design of Issues and Evaluation Scenarios for Multiple Autonomous Driving Systems
- Development of Risk Assessment and Safe Driving Solutions for Hazardous Areas
- Development of Trajectory Optimization Solutions for Autonomous Driving
- Development of a Logistics Automation Solution Using Tow Tractors at Incheon Airport (w/ Korean Air)
- Development of Automation Solutions Utilizing Mobile Robots in Warehouses

Intern, Autonomous Driving Department (IAV GmbH, Germany)

Sep 2019 – Aug 2020

- Development of Lane-change intention detection module using machine learning

QC/QA Engineer (Hyundai Mobis, S.Korea)

Jan 2015 – Mar 2018

- Analysis of ADAS Electronic Components Failures and Derivation of Improvement Measures
- Field Defect Management of ADAS Components in Korea and Europe

Projects

Spot and Atlas from Boston Dynamics (Hyundai Motor Company)

Mar 2025 - Present

- Developed and integrated factory automation solutions using the Boston Dynamics Spot platform, encompassing Safety Patrol, Machine Inspection, and automated Quality Inspection leveraging Spot Arm manipulation.
- Defined the Concept of Operations (ConOps) and technical requirements for deploying humanoid robots in manufacturing environments.

- Humanoid R&D Research: Researched cutting-edge humanoid technologies, including world model integration, simulation, teleoperation, and VLA.
- CES 2026 Exhibition: Demonstrated an automated Quality Inspection solution leveraging Spot Arm manipulation.

Autonomous Mobile Robot (ThorDrive)

Jan 2024 - Mar 2025

- Development of Automation Solutions Utilizing Mobile Robots in Warehouses
 - Period: 1+ years (2024.01 - 2025.03)
 - Result: a multi-agent system embedded in FMS
 - Skills: standard (VDA5050), task allocation (auction), MAPF (CBS, SIPP, ADG), robust execution and replanning (ADG), MQTT, SW testing/verification, CI/CD
 - Tools used: ROS, simulator (MVSIM, Isaac Sim), Eclipse Mosquitto, static & dynamic analysis (clang-tidy, gtest, Valgrind, code coverage), CI/CD (Azure pipeline, Git, Docker)

Autonomous Vehicles (ThorDrive)

Nov 2020 - Mar 2025

- Development of Autonomous Shuttle/Taxi Solutions
 - Period: 1+ years (2020.11 - 2022.04)
 - Result: a jerk-minimized velocity planning module
 - Skills: ST map, MPC, QP, LQR
 - Tools used: ROS, Carla simulator, optimization solver (OSQP), OpenCV
- Development of Automated Parking Path Planning Solutions
 - Period: 6 months (2022.04 - 2022.10)
 - Result: a path planning module for parking service
 - Skills: Reeds-Shepp, Kinodynamic-RRT*, Hybrid A*, QP
 - Tools used: ROS, Carla simulator, OMPL (Open Motion Planning Library)
- Design of Issues and Evaluation Scenarios for Multiple Autonomous Driving Systems
 - Period: 4 months (2022.07 - 2022.11)
 - Result: a paper published in Autumn Annual Conference of IEIE
 - Skills: deadlock scenario generation (double merge scenario, double overtake)
 - Tools used: ROS, Carla simulator
- Development of Risk Assessment and Safe Driving Solutions for Hazardous Areas
 - Period: 10 months (2023.01 - 2023.10)
 - Result: an occlusion-aware risk assessment module, a paper published in RA-L (SCI)
 - Skills: uniform distribution, reachability set
 - Tools used: ROS, Carla simulator, OMPL (Open Motion Planning Library)
- Development of Trajectory Optimization Solutions for Autonomous Driving
 - Period: 1 year (2022.12 - 2023.11)
 - Result: a trajectory optimization module, a paper submitted in T-ITS (SCI, accepted in Oct 2024)
 - Skills: B-spline, swept volume, gradient-based planner, MPC, OCP, QP, L-BFGS
 - Tools used: ROS, Carla simulator, optimization solvers (IPOPT, OSQP, L-BFGS, ACADOS), CasADi
- Development of a Logistics Automation Solution Using Tow Tractors at Incheon Airport (with Korean Air)
 - Period: 1 year (2024.01 - now)
 - Result: successfully completed approximately 10 km of autonomous driving within the airport, carrying the President of Korean Air
 - Skills: car-like tractor kinematics, collision avoidance, intersection handling (signals, unprotected turns)
 - Tools used: ROS, Carla simulator

Autonomous Vehicle (IAV GmbH)

Sep 2019 - Aug 2020

- Development of Lane-change prediction module using machine learning
 - Period: 1 year (2019.09 - 2020.08)
 - Result: Lane-change prediction module, master thesis
 - Skills: HMM, potential field, deep learning (SVM, MLP, LSTM)
 - Tools used: Python (TensorFlow, PyTorch, Scikit-learn), C++ (Eigen, Boost), ADTF

QC/QA (Hyundai Mobis)

Jan 2015 - Mar 2018

- Analysis of ADAS Electronic Components Failures and Derivation of Improvement Measures
 - Period: 7 months (2015.03 - 2015.12)
 - Result: failure Analysis of 15+ Products, including SMK, BCM, MDPS, etc.
 - Skills: CAN, circuit analysis, ASIC swap test
 - Tools used: VECTOR Tools (CANoe, CANape, etc)
- Field Defect Management of ADAS Components in Korea and Europe
 - Period: 2+ years (2016.01 - 2018.03)
 - Result: field monitoring of 30+ components and analysis and improvement of field defect causes for 10+ components
 - Skills: FMEA, 6 sigma

Publications

Safe and Efficient Trajectory Optimization for Autonomous Vehicles using B-spline with Incremental Path Flattening 2024

Jongseo Choi, Hyuntai Chin, Hyunwoo Park, Daehyeok Kwon, Sanghyun Lee, Doosan Baek

[IEEE Transactions on Intelligent Transportation Systems](#) [DOI: 10.1109/TITS.2024.3493060](#)

Occlusion-aware Risk Assessment and Driving Strategy for Autonomous Vehicles Using Simplified Reachability Quantification 2023

Hyunwoo Park, Jongseo Choi, Hyuntai Chin, Sang-Hyun Lee, Doosan Baek

[IEEE Robotics and Automation Letters](#) [DOI: 10.1109/LRA.2023.3329627](#)

The Design of a Test Scenario for Verifying Multi-agent based Edge Connected Urban Autonomous Driving Service 2022

Daehyeok Kwon, Jongseo Choi, Hyuntai Chin, Doosan Baek

[Autumn Annual Conference of IEIE](#)

Lane Change Intention Detection (LCID) of other vehicles using interactive relationships of surrounding multiple objects 2020

Jongseo Choi

Technical University of Chemnitz & IAV GmbH

Development and Evaluation of Lane Change Intention Detection module in ADTF with real-time capability 2019

Jongseo Choi

Technical University of Chemnitz & IAV GmbH

Development of a mini-wheelchair controlled by a wireless helmet using an accelerometer and gyro sensor 2014

Jongseo Choi

Chungbuk National University

Patents

Simulation Apparatus and Method Based on External Trigger Information

2024

Jongseo Choi, Daehyeok Kwon, Hyuntai Chin, Hanbit Kim, Hyuntai Jin, Hyunwoo Park

Korean Patent Application

Publication No.: KR 10-2024-0076866

Application No.: 10-2022-0157995

Publication Date: May 31, 2024

[Google Patents Link](#) 

Technologies

Programming Languages

- C, C++, Python, Java

Technologies

- SW Development Process: Agile, Waterfall, V-Model, ConOps
- SW Testing/Verification: static analysis (clang-tidy, cppcheck), dynamic analysis (gtest, Valgrind, code coverage), MIL, SIL, PIL, HIL
- CI/CD: Azure DevOps pipeline, Git, Docker
- Modularization and OOP: design patterns (e.g., singleton, factory, observer, etc)
- Middleware & Communication: ROS/ROS2 (DDS), gRPC, MQTT, TCP/UDP, Bluetooth, CAN

Miscellaneous Experience

Languages

- Korean (native), English (Professional working proficiency), German (Elementary proficiency)

Awards and Achievements

- Best Employee of the year (2024): ThorDrive
- Merit scholarship (2019): 3,600 Euro from the committee of the Faculty of Computer Science, the president of Chemnitz University of Technology.
- Merit scholarship (2014): 5,000,000 won from scholarship foundation of Taekwang.
- Merit scholarship (2012): 1,000,000 won from Chungbuk National University.

Certificate

- Driver's license 1st class; Regular